

Netflix User Behavior & Revenue Analysis: A SQL Case Study

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Tools Used: SQL | MySQL

Project Objective

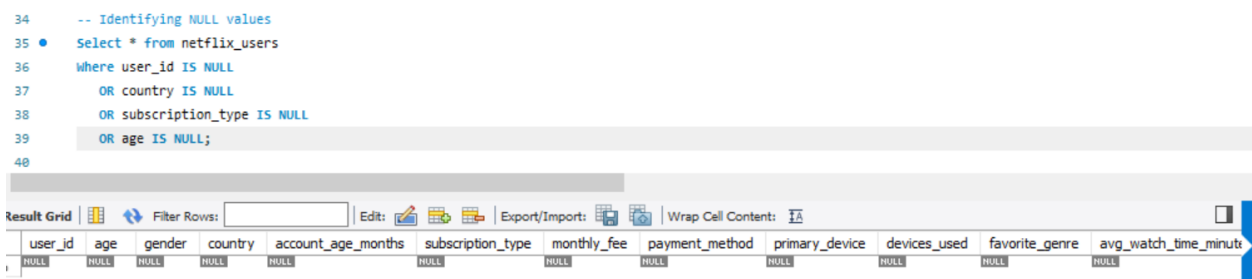
The objective of this project is to analyze Netflix user behavior, subscription trends, customer engagement, and churn patterns using SQL. The analysis focuses on generating business insights related to revenue,

customer preferences, market performance, and user retention.

Data Cleaning & Quality Checks

1) Identifying Missing Values

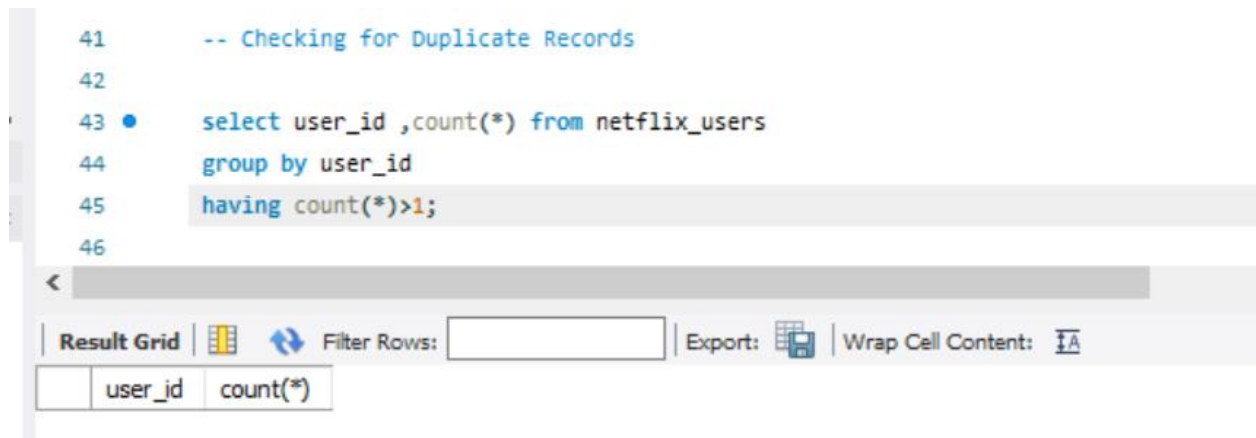
```
34 -- Identifying NULL values
35 • Select * from netflix_users
36 where user_id IS NULL
37    OR country IS NULL
38    OR subscription_type IS NULL
39    OR age IS NULL;
40
```



user_id	age	gender	country	account_age_months	subscription_type	monthly_fee	payment_method	primary_device	devices_used	favorite_genre	avg_watch_time_minut
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

2) Duplicate Record Check

```
41 -- Checking for Duplicate Records
42
43 • select user_id ,count(*) from netflix_users
44    group by user_id
45    having count(*)>1;
46
```



user_id	count(*)
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3) Outlier Check for Age Values

```

47      -- Data Integrity - Outlier Check (Age)
48      •  select count(*) from netflix_users
49      where age < 18 or age > 100;
50

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	count(*)			
▶	0			

4) Monthly Fee Validation Check

```

51      -- - Checking for any records where the monthly fee is zero or negative.
52      •  select * from netflix_users
53      where monthly_fee <= 0;

```

Result Grid		Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
	user_id	age	gender	country	account_age_months
*	NULL	NULL	NULL	NULL	NULL
	subscription_type	monthly_fee	payment_method	primary_device	devices_used
	NULL	NULL	NULL	NULL	NULL
	favorite_genre	avg_watch_time_minut			
	NULL	NULL			

Exploratory Data Analysis Using SQL

Q1 : Total Number of Users.

```

55      -- Q1: Total Number of Users
56      •  select count(*) from netflix_users;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	count(*)			
▶	50000			

Q2: Average Age of Netflix Users.

```
--  
58      -- Q2: Average Age of Netflix Users
```

```
59
```

```
60 • select round(avg(age),0) from netflix_users;
```

```
61
```

```
62      -- Q3: Count of Users by Country
```

<

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	round(avg(age),0)
▶	41

Q3: Count of Users by Country

```
64 • SELECT
65     country, COUNT(*) AS country_wise_users
66 FROM
67     netflix_users
68 GROUP BY country
69 ORDER BY country_wise_users DESC;
```

< Result Grid |   Filter Rows: | Export:

	country	country_wise_users
▶	Brazil	5116
	USA	5087
	India	5028
	Spain	5027
	Germany	5024
	Australia	5004
	Canada	4959
	UK	4929
	France	4919
	Japan	4907

Observation:

Brazil, the USA, and India have the highest number of Netflix users, indicating strong user presence across these markets.

Q4: Distribution of Subscription Types.

```
67      -- Q4: Distribution of Subscription Types
68  ●    select subscription_type,count(*) from netflix_users
69      group by subscription_type;
70
71      -- Q5: Most Popular Genre.
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Cont

subscription_type	count(*)
Standard	19931
Basic	14873
Premium	15196

Observation:

The Standard subscription plan has the highest number of users, indicating strong customer preference for balanced pricing and features.

Q5: Most Popular Genre.

```
71 -- Q5: Most Popular Genre.  
72 • select favorite_genre, count(*) as total_fans from netflix_users  
73 group by favorite_genre  
74 order by total_fans desc;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	favorite_genre	total_fans			
▶	Documentary	6352			
	Romance	6282			
	Comedy	6259			
	Thriller	6257			
	Action	6235			
	Horror	6223			
	Drama	6203			
	Sci-Fi	6189			

Observation:

Documentary emerged as the most popular genre among users, indicating strong audience interest in informative and educational content.

Q6: Gender Distribution by Country.

```

76      -- Q6 Gender Distribution by Country
77      •  select  country, gender, count(*) as count from netflix_users
78          group by country, gender
79          order by country;

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

country	gender	count
Australia	Female	1744
Australia	Male	1604
Australia	Other	1656
Brazil	Female	1732
Brazil	Male	1694
Brazil	Other	1690
Canada	Female	1604
Canada	Male	1676
Canada	Other	1679
France	Female	1657
France	Male	1656
France	Other	1606
Germany	Female	1613
Germany	Male	1748
Germany	Other	1663
India	Female	1681
India	Male	1725

Observation:

Gender distribution varies across countries, highlighting differences in regional audience demographics and user diversity.

Q7: Top 5 Countries with Highest Premium Subscribers.


```

80 -- Q7: Top 5 Countries with highest 'Premium' Subscribers
81 • select country, count(*) as premium_subscribers from netflix_users
82 where subscription_type = 'Premium'
83 group by country
84 order by premium_subscribers desc limit 5;
85

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	country	premium_subscribers				
	USA	1609				
	Brazil	1575				
	Canada	1545				
	India	1537				
	Spain	1515				

Observation:

The USA has the highest number of Premium subscribers, making it one of Netflix's strongest high-value markets.

Q8: Total Monthly Revenue by Subscription Plan

```
91 • SELECT
92     subscription_type,
93     ROUND(SUM(monthly_fee), 2) AS monthly_revenue
94 FROM
95     netflix_users
96 GROUP BY subscription_type;
```

<		
Result Grid  Filter Rows: Export:  Wrap Cell Content: 		
	subscription_type	monthly_revenue
▶	Standard	245835.69
	Basic	183206.27
	Premium	187125.04

Observation:

The Standard subscription plan generates the highest monthly revenue, indicating strong customer preference for balanced pricing and features.

Q9: Most Preferred Payment Method by Country

97	-- Q10: Most Common Payment Method .
98	
99	● SELECT
100	payment_method, COUNT(*) AS user_count
101	FROM
102	netflix_users
103	GROUP BY payment_method
104	ORDER BY user_count DESC;

Result Grid		Filter Rows:	Export:
	payment_method	user_count	
▶	PayPal	12539	
	Credit Card	12502	
	UPI	12496	
	Debit Card	12463	

Observation:

PayPal is the most commonly used payment method among users, followed closely by Credit Card and UPI, indicating strong adoption of digital payment systems across the platform.

Q10: Country with Highest Content Completion Rate.

```

105
106 -- Q10: Country with highest Content Completion Rate.
107
108 • select country, round(avg(completion_rate),2) as avg_completion_rate from netflix_users
109    group by country
110    order by avg_completion_rate desc
111    limit 5;
112

```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	country	avg_completion_rate				
	USA	65.03				
	UK	64.75				
	Spain	64.68				
	Japan	64.67				
	Germany	64.59				

Observation:

The USA has the highest content completion rate, indicating stronger user engagement and consistent viewing behavior among users in this market.

Q11: Subscription Plan with Highest Churn Rate.

```

113 -- Q11 Which subscription type has highest churn Rate users wise and percentage wise ?
114
115 • SELECT
116     subscription_type,
117     COUNT(*) AS total_users,
118     SUM(churned = 'Yes') AS churned_users,
119     ROUND((SUM(churned = 'Yes') * 100) / COUNT(*),
120           2) AS churn_percentage FROM netflix_users
121 GROUP BY subscription_type
122 ORDER BY churn_percentage DESC;
123

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 				
	subscription_type	total_users	churned_users	churn_percentage
▶	Premium	15196	3099	20.39
	Standard	19931	4004	20.09
	Basic	14873	2861	19.24

Observation:

Although the Standard plan has the highest churn count, the Premium plan shows the highest churn percentage, indicating relatively higher cancellation risk among Premium users.

Final Conclusion & Key Insights

- 1) Netflix has a strong global user base of 50,000 users distributed across multiple countries.**
- 2) The average user age is approximately 41 years, indicating strong engagement among mature audiences.**
- 3) The Standard subscription plan generates the highest monthly revenue, making it the top-performing subscription tier.**
- 4) Documentary content emerged as the most popular genre, showing strong audience interest in informative and educational entertainment.**
- 5) Brazil, the USA, and India represent some of Netflix's strongest user markets based on total user count and Premium subscription activity.**
- 6) The USA recorded the highest content completion rate, indicating strong user engagement and viewing consistency.**

7) Although the Standard plan has the highest churn count, the Premium plan shows the highest churn percentage, suggesting relatively higher cancellation risk among Premium users.

This project helped demonstrate how SQL can be used to analyze customer behavior, business performance, engagement trends, and churn patterns to support data-driven decision making.